

VISHWAS KOTHARI

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Education

University of Colorado Boulder Professional Master's in Computer Science — GPA: 3.5/4.0	Aug 2025 – May 2027 Boulder, CO
Vellore Institute of Technology B.Tech in Information Technology — GPA: 8.49/10.0	Aug 2021 – Jun 2025 Vellore, India

Experience

Space Applications Centre – Indian Space Research Organisation (ISRO) <i>Research Intern</i>	Ahmedabad, Gujarat Jan 2025 – Jul 2025
- Built end-to-end data pipelines in Python (SciPy, NumPy, xarray) for processing satellite sensor data, including reprojection, resampling to ~1 km resolution, and pixel-wise NetCDF merging across large-scale datasets. - Developed a Levenberg–Marquardt optimization framework to retrieve bio-optical parameters from spectral reflectance data across the NASA NOMAD dataset (n=105), achieving $R^2 = 0.987$ for spectral reconstruction. - Architected a comprehensive data validation and visualization framework with 6 plot types and 4 statistical metrics (RMSE, MAE, R^2, bias) for cross-sensor quality assessment, enabling data-driven decision-making on satellite-derived products. - Validated chlorophyll-a retrieval accuracy (log-space $R^2 = 0.879$) using MODIS Aqua and OCM imagery, documenting methodology and reproducible workflows in a 60+ page technical report.	

Projects

XAI Credit Lens — Python, SHAP, LIME, DiCE, XGBoost, Optuna, Streamlit — GitHub	2026
- Built an AI-driven explainable credit decisioning framework; Optuna-tuned ensemble (100 trials/model, Stratified 5-Fold CV) achieved 0.782 test AUC on a 30K-record dataset with 9 engineered behavioral features. - Integrated SHAP, LIME, and DiCE counterfactual explanations with automated fairness auditing — zero violations across 4 protected attributes — and regulatory compliance mapping (ECOA, EU AI Act, Fed SR 11-7); deployed interactive Streamlit dashboard for stakeholder communication.	
Financial Document VQA — PyTorch, LayoutLMv3, Donut, Pix2Struct, LoRA, Gradio — GitHub	2026
- Benchmarked 4 vision-language models (Pix2Struct, Donut, LayoutLMv3, OCR+RoBERTa), observing 63–83% performance drop from generic DocVQA to SEC 10-K filings; curated gold-standard benchmark of 79 SEC images / 397 QA pairs across 10 companies. - Domain-specific LoRA fine-tuning on 307 financial QA pairs improved ANLS by 22%, while generic DocVQA fine-tuning on 5,000 examples caused 39% negative transfer; error analysis across 4 question categories identified structural errors (65%) as the primary failure mode.	
XAI-Driven Collision Avoidance for V2V/V2I in IoV — Python, SHAP, LIME, Random Forest	2025
Developed a collision detection system achieving 99.9% AUC on 206K instances with interpretable feature attribution via SHAP and LIME, enabling transparent AI-first safety-critical decision-making.	

Technical Skills

Languages: Python, SQL, Java, C/C++, JavaScript
ML & AI: scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, SHAP, LIME, DiCE, Optuna, LLM workflows
Data Engineering: NumPy, Pandas, SciPy, xarray, NetCDF, ETL Pipelines, Data Validation, Data Visualization
Tools & Cloud: AWS (Certified, 970/1000), Git, Docker, Linux, Streamlit, MySQL, MongoDB, Power BI
Domain Expertise: AI-Driven Analytics, Explainable AI, Fairness Auditing, Regulatory Compliance, Data Governance

Publications & Certifications

Publication: Kothari, V., Krishwanth, B. (2025). Empowering Survivors: Ethical AI for Countering Violence Against Women. *Springer*.
Certification: AWS Certified Cloud Practitioner — Score: 970/1000 (2024)

Leadership & Involvement

Reviewer, *Public Health* and *Social Sciences and Humanities Open* (Elsevier Journals) 2026 – Present
Senior Core Committee Member – E-Cell, VIT-Stellar, VIT-Spartans, Anokha NGO, Youth Red Cross 2023 – 2024